Linear Algebra I: HW6 Partial Solution

Tse-Hung Lin October 21, 2026

If T is a linear transformation between two vector spaces, we denote $\operatorname{im} T := R(T)$, $\ker T := N(T)$.

2-3-13. Let $A = (A_{ij}), B = (B_{ij})$. Then

$$\begin{aligned}\operatorname{tr}(AB) &= \sum_{i=1}^{n}(AB)_{ii} \\ &= \sum_{i=1}^{n}\sum_{k=1}^{n} A_{ik}B_{ki} \\ &= \sum_{k=1}^{n}\sum_{i=1}^{n} B_{ki}A_{ik} \\ &= \sum_{k=1}^{n}(BA)_{kk} = \operatorname{tr}(BA)\end{aligned}$$

and

$$\operatorname{tr}(A^t) = \sum_{i=1}^{n}(A^t)_{ii} = \sum_{i=1}^{n} A_{ii} = \operatorname{tr}(A).$$

2-3-16. (a) Since $\operatorname{im} T^2 \subseteq \operatorname{im} T$, $\operatorname{rank} T^2 = \operatorname{rank} T$ implies $\operatorname{im} T^2 = \operatorname{im} T$. On the other hand, $\ker T \subseteq \ker T^2$, then by the dimension formula

$$\dim V = \dim \operatorname{im} T + \dim \ker T = \dim \operatorname{im} T^2 + \dim \ker T^2.$$

$\ker T = \ker T^2$. If $x \in \operatorname{im} T \cap \ker T$, then $x = T(y)$ for some $y \in V$, and $T(x) = T^2(y) = 0$, thus $y \in \ker T^2 = \ker T$, i.e. $x = T(y) = 0$. Therefore, $\operatorname{im} T \cap \ker T = \{0\}$.

It is clear that $\operatorname{im} T + \ker T$ is a subspace of V, and

$$\dim(\operatorname{im} T + \ker T) = \dim \operatorname{im} T + \dim \ker T - \underbrace{\dim(\operatorname{im} T \cap \ker T)}_{0} = \dim V.$$

Therefore, $V = \operatorname{im} T \oplus \ker T$.

(b) Note that we have a filtration of subspaces

$$\cdots \subseteq \operatorname{im} T^n \subseteq \operatorname{im} T^{n-1} \subseteq \cdots \subseteq \operatorname{im} T^2 \subseteq \operatorname{im} T \subseteq V$$

Which means that we have a chain

$$\cdots \leq \operatorname{rank} T^n \leq \operatorname{rank} T^{n-1} \leq \cdots \leq \operatorname{rank} T^2 \leq \operatorname{rank} T \leq \dim V$$

Since $\dim V$ is finite, there must be $k \in \mathbb{N}$ such that $\operatorname{rank} T^k = \operatorname{rank} T^{k+1}$. Then by using the argument above, we can prove that $\operatorname{im} T^k = \operatorname{im} T^{k+1}$, $\ker T^k = \ker T^{k+1}$, and that $V = \operatorname{im} T^k \oplus \ker T^k$.

2-3-17. We claim that such linear transformation must be a projection: Suppose that $T^2 = T$, then decompose $v \in V$ into

$$v = T(v) + (v - T(v))$$

Since $T(v-T(v)) = T(v)-T(T(v)) = 0$, $v-T(v) \in \ker T$, we have $V = \operatorname{im} T + \ker T$. If suffices to prove that $\operatorname{im} T \cap \ker T = \{0\}$, suppose that $x \in \operatorname{im} T \cap \ker T$, then $x = T(y)$ for some $y \in V$ and $T(x) = 0$, therefore,

$$0 = T(x) = T(T(y)) = T(y) = x,$$

Therefore, we have $V = \ker T \oplus \operatorname{im} T$. Given $v \in V$, we can decompose $v = w_1 + w_2$ uniquely where $w_1 \in \ker T$ and $w_2 \in \operatorname{im} T$, then $T(v) = T(w_1) + T(w_2) = T(w_2)$, since $T^2(v) = T(v)$, we have $T(w_2) = w_2$. That is $T(v) = w_2$, so T is a projection on $\operatorname{im} T$ along $\ker T$.

2-4-1. FTFFTFTTT

2-4-2. NNYNNY

2-4-4. Since A, B are invertible, A^{-1}, B^{-1} exist and one can check that $AB(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})AB$, by the uniqueness of inverse, $(AB)^{-1} = B^{-1}A^{-1}$.

2-4-5. It suffices to show that $(A^{-1})^t A^t = I = A^t(A^{-1})^t$. Recall that $A^tB^t = (BA)^t$ for all square matrices A, B. Thus,

$$(A^{-1})^t A^t = (AA^{-1})^t = I^t = I = I^t = (A^{-1}A)^t = A^t(A^{-1})^t.$$

2-4-9. Since AB is invertible, the linear map

$$L_{AB} : F^n \longrightarrow F^n \qquad v \longmapsto ABv$$

is an isomorphism. Furthermore, $L_{AB} = L_A \circ L_B$, so L_B is one-to-one, and L_A is onto, but since L_A, L_B are linear maps between vector space with the same finite dimension, L_A, L_B are both isomorphisms. Therefore, $[L_A]_\beta = A, [L_B]_\beta = B$ are invertible matrices (here β is the standard basis for F^n).

2-4-10. (a) Since $AB = I_n$ is invertible, A, B are invertible.

(b) Since A is invertible, $AB = I \implies ABA = A \implies BA = I$, thus B is the inverse of A. Similarly, A is the inverse of B.

(c) WE prove the following

Proposition. *Let V be an n-dimensional vector space and $T, U : V \to V$ be linear such that $TU = \mathrm{id}_V$, then T, U are isomorphisms and $T = U^{-1}, U = T^{-1}$.*

Proof. Fix an order basis β for V, then $[TU]_\beta = [\mathrm{id}_V]_\beta = I_n$. On the other hand, $[TU]_\beta = [T]_\beta[U]_\beta = I_n$, thus $[T]_\beta, [U]_\beta$ are invertible matrices, which implies T, U are isomorphisms. □

2-4-14. An example of isomorphism would be

$$T : V \longrightarrow F^3 \qquad \begin{pmatrix} a & a+b \\ 0 & c \end{pmatrix} \longmapsto (a, a+b, c).$$

2-4-15. If T is an isomorphism, it is 1-1, so $T(\beta)$ is linearly independent (since β is linearly independent). Furthermore, since T is also onto,

$$W = \operatorname{im} T = T(V) = T(\operatorname{span}(\beta)) = \operatorname{span} T(\beta).$$

Therefore, $T(\beta)$ generates W. $T(\beta)$ is a basis.

Conversely, if $T(\beta)$ is a basis for W, then T is 1-1. Furthermore,

$$W = \operatorname{span} T(\beta) = T(\operatorname{span}(\beta)) = T(V) = \operatorname{im} T,$$

so T is also onto.

2-4-16. Given $X, Y \in M_{n\times n}(F), c \in F$, we have

$$\Phi(X+cY) = B^{-1}(X+cY)B = B^{-1}(XB+cYB) = B^{-1}XB+cB^{-1}YB = \Phi(X)+c\Phi(Y).$$

Φ is linear. Suppose $X \in \ker \Phi$, then $B^{-1}XB = O \implies X = O$. Φ is 1-1. Given $Y \in M_{n\times n}(F)$, define $Z = BYB^{-1} \in M_{n\times n}(F)$, then $\Phi(Z) = B^{-1}(BYB^{-1})B = Y$, thus Φ is onto. Therefore, Φ is an isomorphism.

2-4-17. (a) Check subspace axioms carefully. (Note that this holds for any linear transformation, not just for isomorphisms).

(b) Since V is finite dimensional, V_0 is also finite dimensional, let β be a basis for V_0, then $T(\beta)$ is linear independent since T is 1-1, and $T(V_0) = T(\operatorname{span}(\beta)) = \operatorname{span} T(\beta)$, thus $T(\beta)$ is a basis for $T(V_0)$, i.e. $\dim T(V_0) = |T(\beta_0)| = |\beta_0| = \dim V_0$. (Note that we only use the injectivity of T, so it holds for any 1-1 linear map on finite dimensional vector spaces).

2-4-20. Fix an ordered basis β for V, γ for V we have a commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{\quad T \quad} & W \\ {\scriptstyle \phi_\beta}\downarrow\wr & & {\scriptstyle \phi_\gamma}\downarrow\wr \\ F^n & \xrightarrow[L_A, A=[T]^\gamma_\beta]{} & F^m \end{array}$$

Then

$$\begin{aligned} \operatorname{rank} L_A = \dim(\operatorname{im} L_A) &= \dim(L_A(F^n)) \\ &\overset{\phi_\beta \text{ onto}}{=} \dim L_A(\phi_\beta(V)) \\ &\overset{\text{diagram commutes}}{=} \dim \phi_\gamma T(V) \\ &\overset{2\text{–}4\text{–}17.}{=} \dim T(V) = \operatorname{rank} T \end{aligned}$$

Consequently, since V is finite dimensional, by the dimension formula

$$\begin{aligned} \text{nullity } L_A &= \dim F^n - \operatorname{rank} L_A \\ &= \dim V - \operatorname{rank} T \\ &= \text{nullity } T. \end{aligned}$$

2-4-22. Given $f, g \in P_n(F)$, $t \in F$,

$$\begin{aligned} T(f+tg) &= ((f+tg)(c_0), \cdots, (f+tg)(c_n)) \\ &= (f(c_0)+tg(c_0), \cdots, f(c_n)+tg(c_n)) \\ &= (f(c_0), \cdots, f(c_n)) + t(g(c_0), \cdots, g(c_n)) = T(f) + tT(g) \end{aligned}$$

T is linear. Suppose $T(f) = T(g)$ for some $f, g \in P_n(F)$, then $T(f, g) = 0$. That is, $c_0, \cdots, c_n$ are roots of $f - g$. However if $f - g$ is nonzero, then there must be at most n roots, thus $f - g = 0$ is the zero polynomial, that is, $f = g$, so T is 1-1. It suffices to prove that T is onto:

Given $(a_0, a_1, \cdots, a_n) \in F^{n+1}$, define

$$f(x) = \sum_{i=0}^{n} a_i \prod_{0 \le j \ne i \le n} \frac{(x - c_j)}{(c_i - c_j)} \in P_n(F).$$

Then $T(f) = (a_0, \cdots, a_n)$. Thus, T is an isomorphism.

2-5-4. $[T]_\beta = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}, [T]_{\beta'} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 13 \\ -5 & 9 \end{pmatrix}$

2-5-5. $[T]_\beta = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, [T]_{\beta'} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ 1/2 & -1/2 \end{pmatrix}$

2-5-6. (a) $[L_A]_\beta = \begin{pmatrix} 6 & 11 \\ -2 & -4 \end{pmatrix}, Q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

(b) $[L_A]_\beta = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}, Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

(c) $[L_A]_\beta = \begin{pmatrix} 2 & 2 & 2 \\ -2 & -3 & -4 \\ 1 & 1 & 2 \end{pmatrix}, Q = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix}$

(d) $[L_A]_\beta = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 18 \end{pmatrix}, Q = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -2 & 0 & 1 \end{pmatrix}$. The is the diagonalization!!

4. This is Theorem 2.18.

5. $(i) \implies (ii)$ from definition. $(ii) \implies (iii)$ from Theorem 2.5. $(iii) \implies (i)$ from Theorem 2.5 and definition.